

Machine Learning for Aquatic Ecotoxicity

Prediction with Quantified Uncertainty

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Abstract

Aquatic ecosystems face increasing pressure from the thousands of industrial chemicals entering rivers and lakes annually, yet our ability to experimentally assess the risks of these compounds lags significantly behind their production. To protect these environments, regulators rely on toxicity data—specifically Lethal Concentration 50% (LC50) values—to determine safe thresholds. However, the global matrix of potential chemical-species interactions is overwhelmingly sparse, with over 99.5% of possible combinations untested. While machine learning techniques, particularly Factorization Machines (FM), have shown promise in “filling these gaps” by predicting missing toxicity values, they typically provide only point estimates. In a domain where safety is paramount, a single predicted number is insufficient; regulators and risk assessors require a measure of confidence to make informed decisions.

This article presents a robust solution to the sparsity problem: the Bayesian Factorization Machine (BFM). By re-framing the model parameters as probability distributions rather than fixed constants, the BFM not only achieves state-of-the-art predictive accuracy—matching

the noise floor of inter-test variability—but also provides a rigorous quantification of uncertainty. Crucially, the model decomposes this uncertainty into two distinct components: aleatoric uncertainty, reflecting the inherent noise and variability in biological testing, and epistemic uncertainty, reflecting the model’s lack of knowledge in unexplored areas of the data.

We demonstrate how this quantified uncertainty transforms downstream risk assessment workflows. First, we show how BFM predictions can be used to construct Species Sensitivity Distributions (SSDs) that include credible intervals and posterior distributions for hazardous concentration thresholds (HC5), moving beyond static curves to dynamic, uncertainty-aware risk profiles.

Furthermore, we explore the practical utility of epistemic uncertainty through a simulation of Active Learning. We illustrate how an "uncertainty subsampling" strategy can leverage the BFM’s outputs to intelligently prioritize future laboratory tests. By targeting experiments that maximize information gain, this approach can achieve target accuracy levels with significantly fewer samples than random testing, offering a pathway to more cost-effective and efficient environmental protection.

Dataset

The dataset used throughout this thesis is the ADORE dataset introduced by ?, which is a dataset designed for fostering the adoption of Machine Learning in ecotoxicology. The ecotoxicity values in ADORE come from the US EPA ECOTOX dataset[?], and ADORE further includes a large number of features of chemicals and species forming a low barrier to entry for data scientists from outside the domain, and reducing the risk of errors when procuring features. Multiple endpoints are available, such as Effective Concentration or No Effect Concentration, however out of the available endpoints, LC50 is chosen as the sole endpoint to focus on for all experiments, as it is consistently well-defined and has a significant

amount of data available. Future work could work on using data from multiple endpoints to improve model performance, however this would require significant data engineering work to standardise the reporting of other endpoints.

Each row of the dataset contains an LC50 observation, along with information relating to the chemical that was tested, the species that was exposed, and further additional information about the lab conditions for some observations. The high feature-richness of this dataset is what makes it an attractive choice for experimenting with Machine Learning, especially for practitioners outside of the ecotox domain.

The ADORE dataset includes a number of 'challenge' slices, where the data is limited to specific sub-species, however, in order to leverage as much of the data as possible, the processed but not divided dataset is used throughout this thesis. Specifically `ecotox_mortality_processed.csv` is joined with `ecotox_properties_with-oecd-function.csv` available to download at <https://gitlab.renkulab.io/mltox/adore>, where in-depth detail on the dataset processing can also be found.

Quantitative properties of the dataset

Table 1: Summary statistics of the ADORE dataset

Statistic	Value
Number of unique chemicals	3,295
Number of unique species	1267
Number of durations	4
Number of unique (species, chemical, duration) triplets observed	31,319
Total number of observations	70,670
Sparsity (species*chemical matrix)	99.5%

Table 1 shows summary statistics about the dataset, and elucidates the first major challenge of this dataset - **sparsity**. Even if the duration dimension is collapsed, only around 0.05% of the chemical-species pairs have observed LC50 values. This feature of the dataset informs further modelling choices, and highlights the inherent difficulty of the problem. Even filling this matrix is a major modelling challenge, and the amount of data is truly miniscule

when the size of the chemical universe is considered (around 350,000 chemicals are in active use in europe[?]).

The second major challenge is the **uneven distribution of observations** across the matrix, which is illustrated in Figure 1 from the point of view of chemicals, and species separately. The rank-frequency plots illustrate that most unique species and chemicals have just a handful of observations, while a few chemicals and species have been tested hundreds of times.

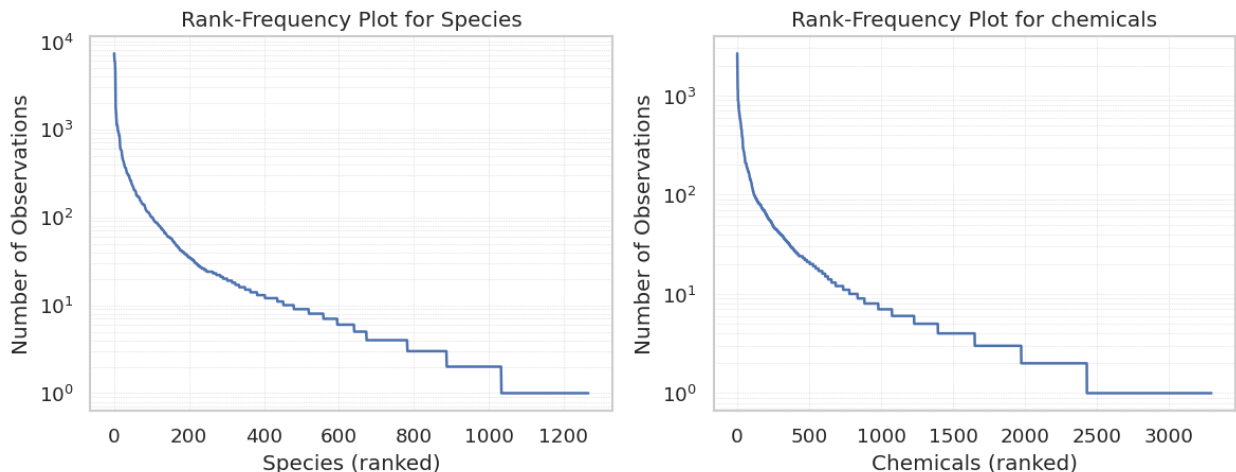


Figure 1: Plots showing the number of observations per species (left) and per chemical (right) ranked by number of observations.

The third challenge of the data is very high inter-test variability. Figure 2 shows the distribution of normalized standard deviations for chemical*species*durations where at least 10 observations are available. This shows that there is a significant portion of the dataset where the variance of repeated observations is very high, more than the mean or even double or triple the mean.

These three challenges are not unique to the ADORE data, but a common issue in ecotox datasets, and these qualities drive model choice and evaluation approaches in applying machine learning in this domain.

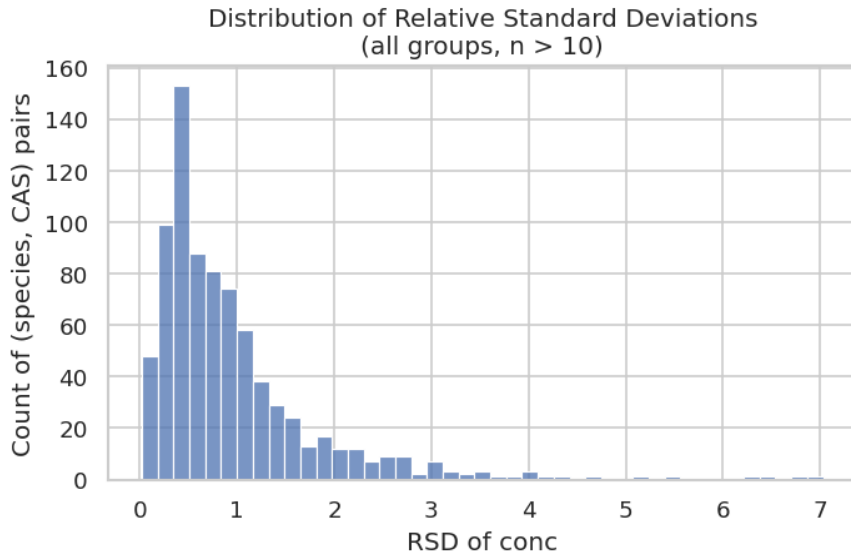


Figure 2: Distribution of RSD’s in the data grouped by species, chemical, duration

Methods

Hierarchical Bayesian Factorization Machine

We select the Factorization Machine (FM) as our foundational architecture due to its proven state-of-the-art performance on the ADORE dataset and its specific efficacy in handling high-dimensional, sparse ecotoxicological data. However, standard predictive implementations—such as basic Gaussian process regression or neural networks—typically rely on a homoscedastic noise assumption, learning a single global variance parameter. This approach is limiting in the context of ecotoxicity, as it conflates the inherent variability of well-studied, stable chemicals with that of poorly characterized or volatile ones, leading to poorly calibrated confidence intervals.

To address this limitation, we introduce the Hierarchical Bayesian Factorization Machine (HierBFM). Adapted from the implementation by Rendle (2012), this model extends the classical framework by incorporating a per-chemical (group-specific) noise precision parameter regularized by a shared hierarchical prior. This formulation allows the model to: (1) learn chemical-specific aleatoric uncertainty, distinguishing between inherently noisy and consis-

tent data; (2) achieve shrinkage towards a global mean, allowing rare chemicals to borrow statistical strength from the population; and (3) explicitly decompose predictive uncertainty into irreducible aleatoric noise and reducible epistemic uncertainty.

Full Model Definition

Let the dataset consist of N observations. Each observation i is characterized by: $\mathbf{x}_i \in R^p$, a sparse feature vector encoding chemical identity, species identity, test duration, taxonomic family, taxonomic class, and numerical chemical properties; $y_i \in R$, the observed log-transformed toxicity value; and $c_i \in \{1, \dots, C\}$, the index of the chemical (group needed for noise term α_c) associated with observation i .

The response y_i is modeled as a noisy observation from a factorization machine:

$$y_i \mid \mathbf{x}_i, \theta, \alpha_{c_i} \sim \mathcal{N}(\hat{y}_i(\theta), \alpha_{c_i}^{-1}) \quad (1)$$

where $\alpha_c > 0$ is the *precision* (inverse variance) specific to chemical c , and the predictive mean $\hat{y}_i(\theta)$ is given by a second-order factorization machine:

$$\hat{y}_i(\theta) = w_0 + \sum_{j=1}^p w_j x_{ij} + \sum_{j=1}^p \sum_{j' > j}^p \langle \mathbf{v}_j, \mathbf{v}_{j'} \rangle x_{ij} x_{ij'} \quad (2)$$

where $w_0 \in R$ is the global bias (intercept), $\mathbf{w} = (w_1, \dots, w_p)^\top \in R^p$ are the linear (main effect) weights, $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_p)^\top \in R^{p \times k}$ is the matrix of latent factor vectors with each $\mathbf{v}_j \in R^k$, and $\langle \mathbf{v}_j, \mathbf{v}_{j'} \rangle = \sum_{f=1}^k v_{jf} v_{j'f}$ is the inner product capturing pairwise interactions.

Hierarchical Prior Structure.

The global bias is assigned a Gaussian prior:

$$w_0 \sim \mathcal{N}(\mu_0, \gamma_0^{-1}) \quad (3)$$

with $\mu_0 = 0$ and $\gamma_0 = 1$. The linear weights are modeled with a shared Normal-Gamma prior inducing adaptive shrinkage:

$$w_j \mid \mu_w, \lambda_w \sim \mathcal{N}(\mu_w, \lambda_w^{-1}) \quad \text{for } j = 1, \dots, p \quad (4)$$

$$\mu_w \mid \lambda_w, \mathbf{w} \sim \mathcal{N}(\bar{w}_{\text{post}}, (\gamma_0 + p\lambda_w)^{-1}) \quad (5)$$

$$\lambda_w \mid \mathbf{w}, \mu_w \sim \text{Gamma}\left(a_\lambda + \frac{p}{2}, b_\lambda + \frac{1}{2} \sum_{j=1}^p (w_j - \mu_w)^2\right) \quad (6)$$

with weakly informative hyperpriors $a_\lambda = b_\lambda = 1$. Analogously, the latent factors share a Normal-Gamma prior:

$$v_{jf} \mid \mu_v, \lambda_v \sim \mathcal{N}(\mu_v, \lambda_v^{-1}) \quad \text{for all } j, f \quad (7)$$

$$\mu_v \mid \lambda_v, \mathbf{V} \sim \mathcal{N}(\bar{v}_{\text{post}}, (\gamma_0 + pk\lambda_v)^{-1}) \quad (8)$$

$$\lambda_v \mid \mathbf{V}, \mu_v \sim \text{Gamma}\left(a_\lambda + \frac{pk}{2}, b_\lambda + \frac{1}{2} \sum_{j,f} (v_{jf} - \mu_v)^2\right) \quad (9)$$

Extension to Hierarchical Per-Chemical Precision.

Each chemical's precision is drawn from a shared Gamma prior:

$$\alpha_c \sim \text{Gamma}(a_0, b_0) \quad \text{for } c = 1, \dots, C \quad (10)$$

where $a_0 = b_0 = 1$ (an exponential prior with mean 1). Given the observations, the posterior for each α_c is conjugate:

$$\alpha_c \mid \{y_i, \hat{y}_i : c_i = c\} \sim \text{Gamma}\left(a_0 + \frac{N_c}{2}, b_0 + \frac{1}{2} \sum_{i:c_i=c} (y_i - \hat{y}_i)^2\right) \quad (11)$$

where $N_c = |\{i : c_i = c\}|$ is the number of observations for chemical c . Chemicals with many observations will have posteriors dominated by the data's sum of squared errors, while chemicals with few observations will have posteriors strongly influenced by the prior, achieving

partial pooling.

Inference via Gibbs Sampling

Posterior inference is performed using a Gibbs sampler, which iteratively samples each parameter from its full conditional distribution. For full details see `rendle` or the code.

The sampler is run for T total iterations, with the first T_{burn} samples discarded as burn-in.

Prediction and Uncertainty Quantification

Posterior Predictive Mean. For a new observation with features $\mathbf{x}^*$ belonging to chemical c^* , the posterior predictive mean is:

$$\hat{y}^* = \frac{1}{S} \sum_{s=1}^S \hat{y}^{(s)}(\mathbf{x}^*) \quad (12)$$

where $\hat{y}^{(s)}(\mathbf{x}^*)$ is the prediction from the s -th posterior sample and S is the number of post-burn-in samples.

Epistemic Uncertainty. Epistemic uncertainty reflects uncertainty in the model parameters due to limited training data:

$$\text{Var}_{\text{epistemic}}(\mathbf{x}^*) = \frac{1}{S} \sum_{s=1}^S \left(\hat{y}^{(s)}(\mathbf{x}^*) - \hat{y}^* \right)^2 \quad (13)$$

This uncertainty is reducible—it decreases as more data becomes available.

Aleatoric Uncertainty. Aleatoric uncertainty reflects inherent variability in the data-generating process:

$$\text{Var}_{\text{aleatoric}}(c^*) = \frac{1}{S} \sum_{s=1}^S \frac{1}{\alpha_{c^*}^{(s)}} \quad (14)$$

where $\alpha_{c^*}^{(s)}$ is the s -th posterior sample of the precision for chemical c^* .

Total Predictive Variance. The total predictive variance combines both sources:

$$\text{Var}_{\text{total}}(\mathbf{x}^*, c^*) = \text{Var}_{\text{epistemic}}(\mathbf{x}^*) + \text{Var}_{\text{aleatoric}}(c^*) \quad (15)$$

SSD construction

The novel SSD format introduced in Posthuma et al. 2025 is used. This idea is extended to use the uncertainty around individual predictions to produce entire SSD curves. There are multiple ways in which such an uncertainty SSD can be constructed, the most straightforwardly robust method used in this work is monte-carlo sampling from the predicted distribution for each species. The exact procedure is described as follows.

To construct Species Sensitivity Distributions (SSDs) with uncertainty bands, we leveraged the full predictive distribution from the hierarchical Bayesian Factorization Machine. For a given chemical at a specified exposure duration, the model provides a predicted toxicity (log LC50) and associated uncertainty for each of the 1,267 species in the dataset, including species that were not empirically tested with that chemical. The total predictive uncertainty for each chemical-species pair was computed as the sum of epistemic variance (derived from the variance across posterior Gibbs samples) and aleatoric variance (the chemical-specific noise term learned by the hierarchical model). We employed Monte Carlo sampling to propagate these uncertainties into the SSD: for each of 1,000 iterations, a toxicity value was independently sampled from the normal predictive distribution $\mathcal{N}(\mu, \sigma)$ for each species, where μ is the predicted mean toxicity and σ is the total predictive standard deviation. Each set of 1,267 sampled toxicity values was then sorted to produce one realization of the species sensitivity ranking. Across all Monte Carlo samples, we computed the median SSD curve along with 50% and 99% credible intervals at each rank position by taking the corresponding percentiles. The hazardous concentration protecting 95% of species (HC5) was estimated from the 5th percentile rank position, with uncertainty bounds derived from the 0.5th and 99.5th percentiles of the Monte Carlo distribution at that rank.

Test prioritization

NOT DONE YET: The idea here is to show how the uncertainty quantification from the model could in theory be used for test prioritization. This is an analogy to active learning in AI. It's quite easy to demonstrate in principle: we could prioritize testing that would decrease model uncertainty the most, in theory using fewer test to achieve confident estimates. This is easy to demonstrate but would be perhaps quite hard to message convincingly around it.

Results

Predictive accuracy

For $k=32$, $N_{iter}=200$, $n_{burn}=100$ the RMSE is 0.76 (log mg/L), which is in line with strong performance on this view of ADORE. Further tuning of the predictive accuracy is likely possible through careful prior tuning or longer training, but this work focuses on downstream usecases of the model.

3 shows a correlation plot of the observations for a visual impression of the predictions.

A further inspection of the residuals 4 allows for a deeper understanding of the biases of the model. In general the predictions are aligned with measurement, however, at low toxicity values the model over predicts, and under-predicts for higher toxicity values, showing a regression to the mean effect (the model 'playing it safe' with predictions).

Uncertainty calibration

- some visualisations of interest of the uncertainty predictions, simple statistics, ratio, plots etc.

Around 15% of total variance is epistemic, the rest is predicted by the model to be aleatoric. It is likely that some part of the epistemic uncertainty is leaking into the aleatoric,

The model tends to overestimate aleatoric uncertainty, however an interesting artifact is that the empirical standard deviation is higher as more observations per triplet are in the

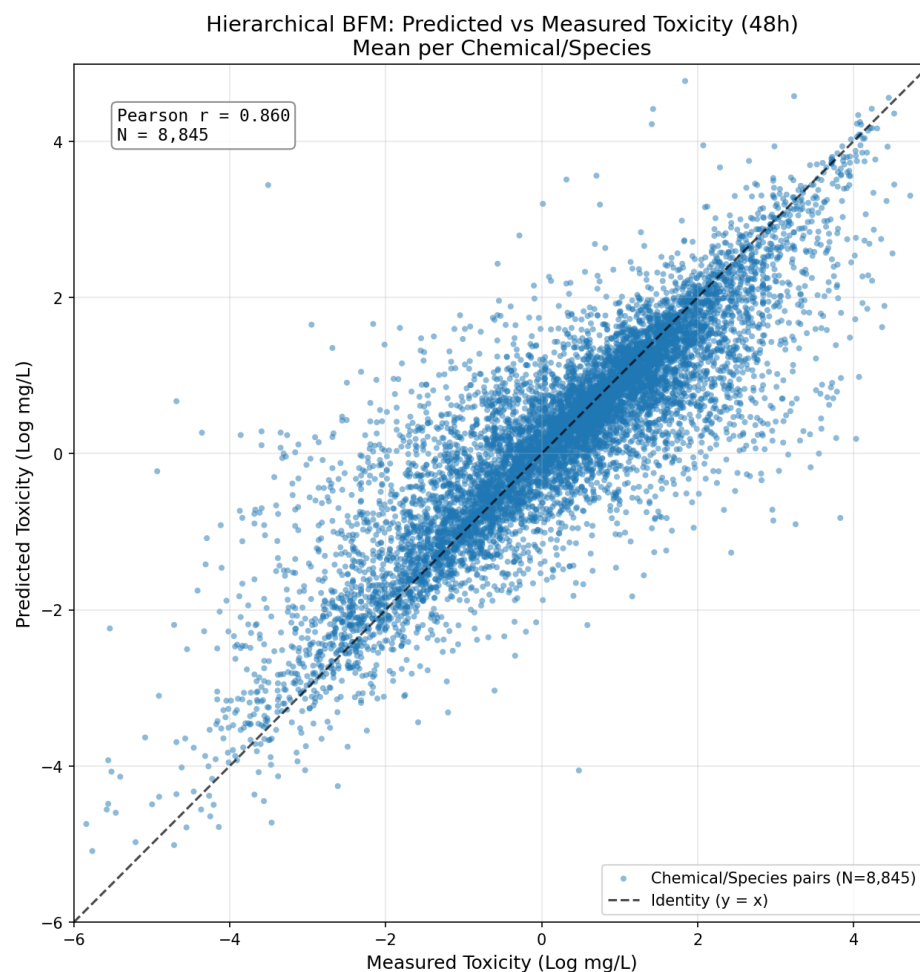


Figure 3: Predictions versus measured (average where multiple measurements available) toxicity for species chemical pairs at 48hrs exposure, correlation plot.

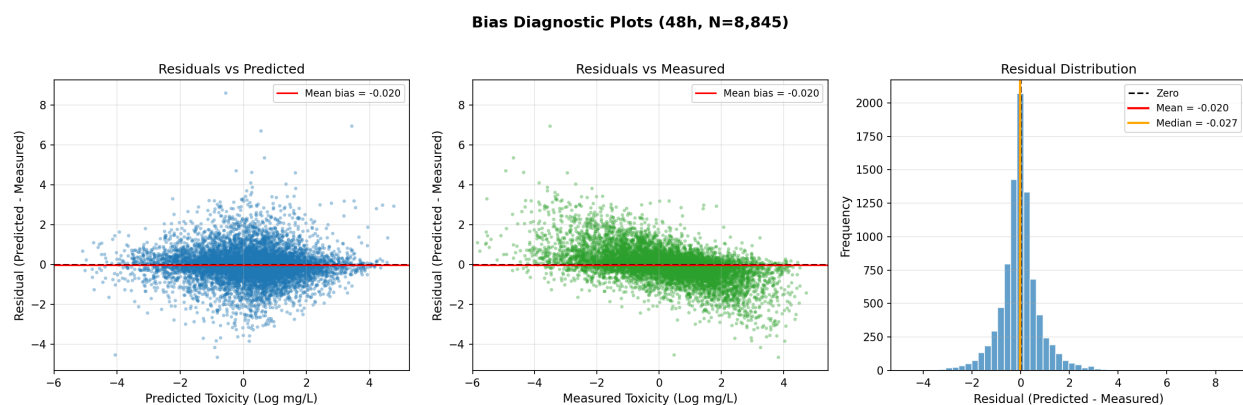


Figure 4: Enter Caption

Table 2: For triplets with repeated measurements, the empirical standard deviation is compared to the predicted SD (aleatoric uncertainty).

Replicates	N groups	Median Ratio	Mean Pred SD	Mean Emp SD
5–9	1,803	1.87	0.61	0.39
10–19	658	1.74	0.55	0.38
20–49	227	1.48	0.53	0.40
50–99	30	1.16	0.54	0.46
100+	6	0.96	0.51	0.56

dataset. The model’s predicted uncertainty is the most calibrated to the triplets with large numbers of observations. This could be artifacts of the dataset, or expected behaviour of the model learning the SD’s that satisfy the largest number of loss points.

SSD’s

FIGURE HERE SUMMARIZING THE PIPELINE (graphical abstract-style)

Recap on SSD’s. We will refer to the following mathematically distinct plots as SSDs:

- Traditional SSD - a curve fitted to actual lab measured observations from the dataset
- Novel SSD - a scatter plot of model predicted mean toxicity for a specific chemical and duration, ordered by toxicity, yielding the familiar SSD shape (following Posthuma et al 2025)
- Novel SSD with Uncertainty Bands - an extension of the novel SSD enabled by the uncertainty predictions of the BFM created using the monte carlo sampling process explained earlier.

5 shows a traditional SSD for Atrazine at 48 hour exposure, fitted to the observations in the ADORE dataset. 6 shows the analogous novel format. 7 shows the novel and traditional SSDs for Atrazine on the same panel (7a, and next to it shows the same comparison for chlorfenprop methyl, which was selected for it’s small number of test results in the dataset, and shows a much higher discrepancy from the fitted SSD (7b).

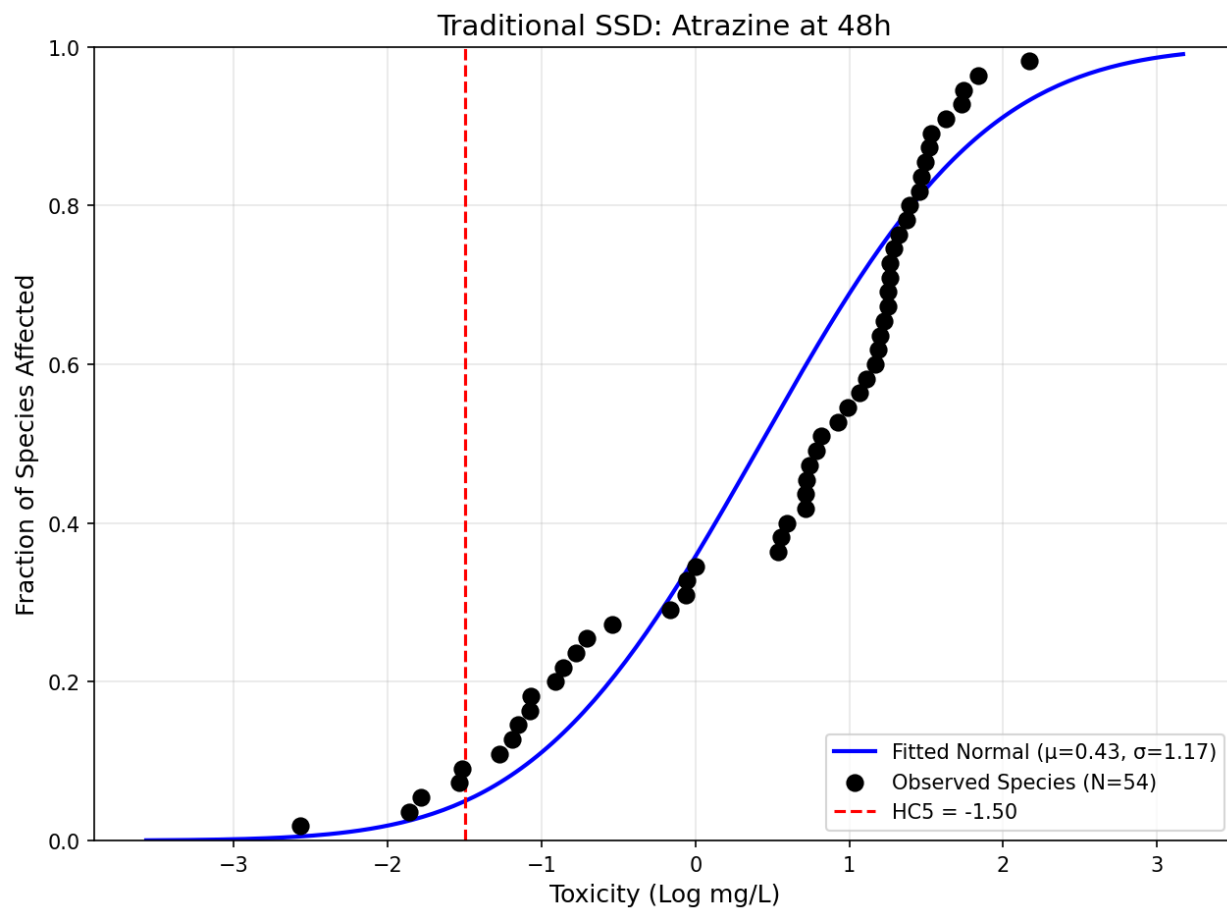


Figure 5: Traditional SSD fitted to real observations (black dots) in the dataset for Atrazine at 48hr exposure.

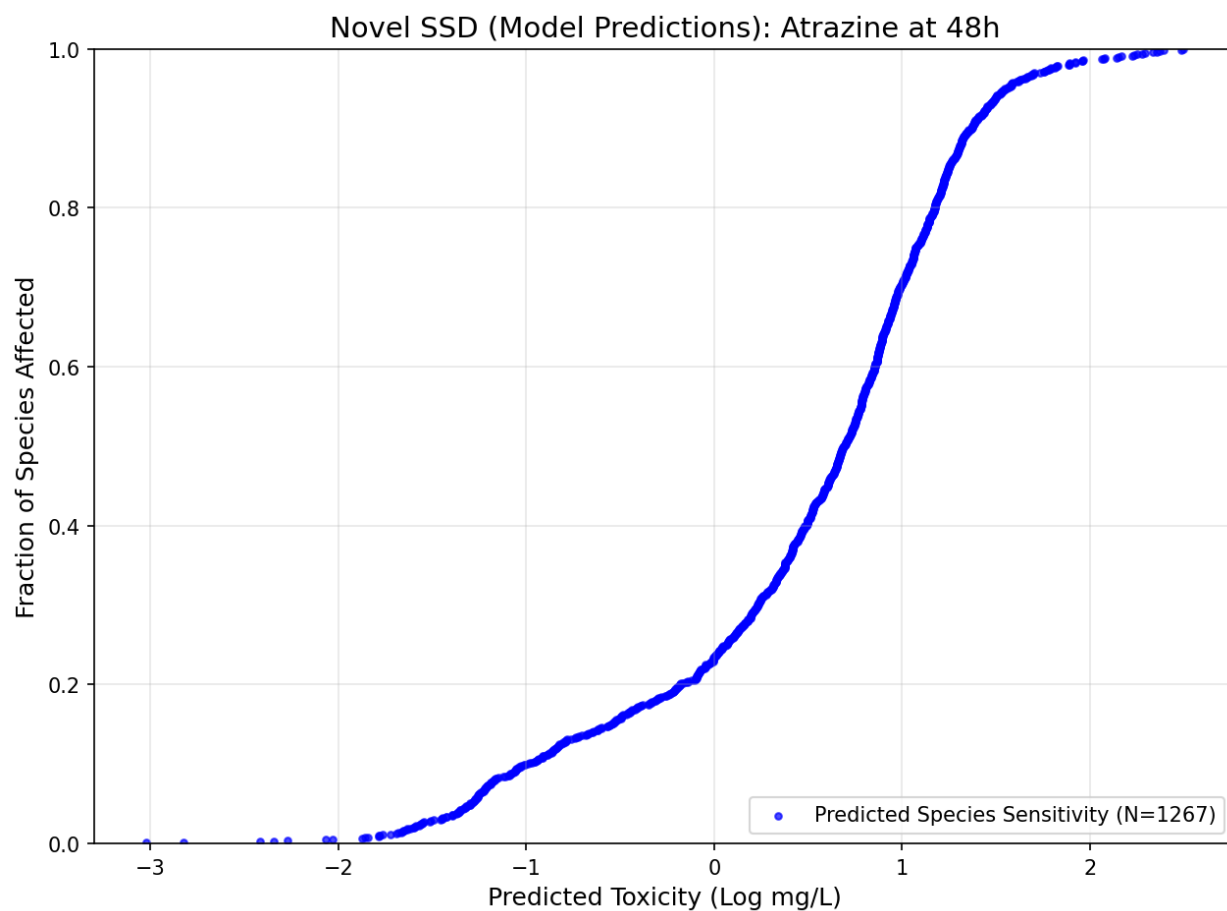
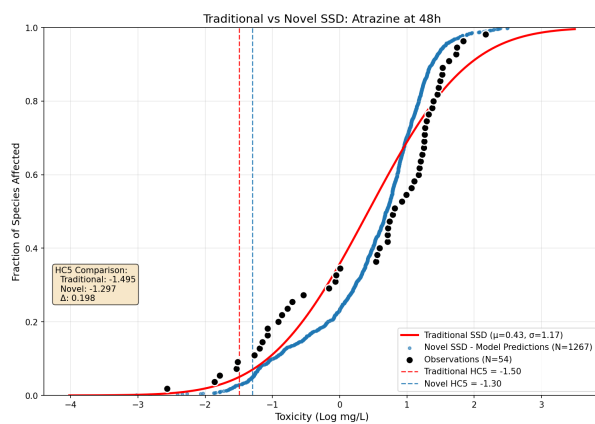
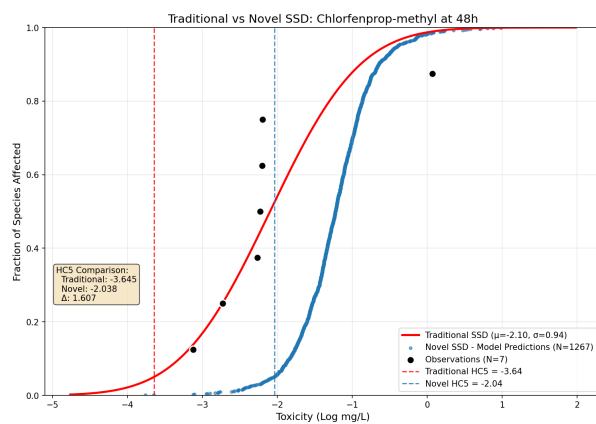


Figure 6: Novel SSD for Atrazine 48hrs. Raw BFM model predicted means for toxicity for each species sorted and scattered.



(a) Atrazine 48hrs.



(b) Chlorfenprop methyl 48hrs.

Figure 7: Side-by-side comparison of Novel vs. Traditional SSDs for two different compounds.

SSD's with uncertainty

The Bayesian Factorization Machine Model predicts a variance term for each chemical in addition to the mean toxicity prediction for each chemical/species pair, which is intended to capture the biological and experimental variance based on the dataset on a per-chemical granularity. Additionally for each prediction, the variance of the predicted means from each model in the sample shows the modelling uncertainty.

The plots below show the Novel SSD plot for Atrazine and Chlorfenprop-methyl where for each plotted prediction, it's two associated uncertainty metrics are shown. Once again Atrazine is selected as a commonly tested chemical, while chlorfenprop methyl is a sparse chemical, having measurements for only 7 unique species (vs 144 for Atrazine).

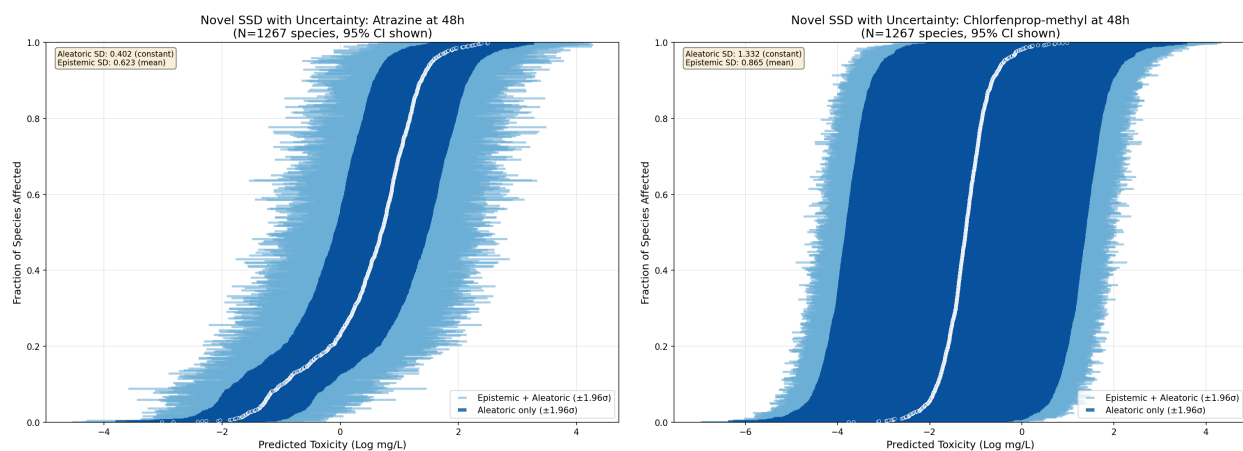


Figure 8: Novel scatter-SSDs for Atrazine (left) and Chlorfenprop (right) with uncertainty bars added for the uncertainty around model predictions for each point. Dark blue is the aleatoric uncertainty, which is constant for each chemical and predicted by the model. Light blue shows the additional epistemic uncertainty, which is different for each species and chemical.

The comparison side-by-side shows how a less tested compound has much higher uncertainty around the predictions, and it is noteworthy the difference in aleatoric (constant, predicted by the model) uncertainty is much greater (around 300%) than the difference in epistemic (modelling) uncertainty (around 40% greater).

Adding the uncertainty bars to the SSD plots in this way is a nice way to visualise the uncertainty around each point, but it should **not be interpreted as the uncertainty**

of the SSD, it shows the uncertainty around each point. The SSD itself is far less uncertain as an order statistic. Intuitively, because it doesn't matter which species is at which location, only the sensitivity at each rank, a lot of the uncertainty will cancel out. With the monte-carlo sampling process we can generate the below SSD's:

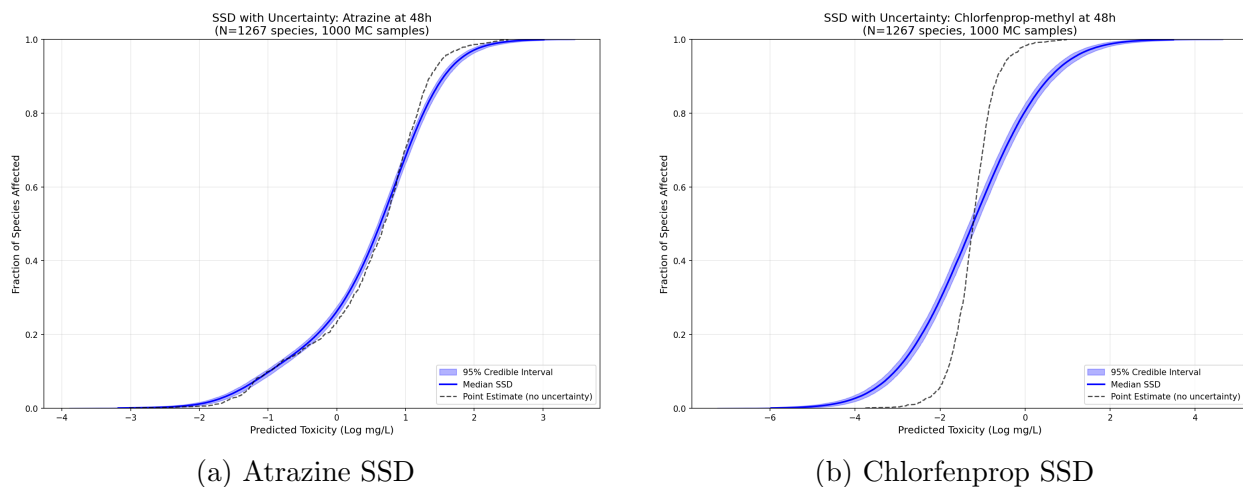


Figure 9: Comparison of SSD results for Atrazine and Chlorfenprop

HC5 Comparison

A more systematic view of the model's predictions and uncertainty can be gained from focusing on the HC5 values. ?? shows a correlation plot between the HC5 values obtained from the traditional curve fitting approach, and the HC5 values read from the uncertainty SSD's from monte-carlo sampling.

The correlation plots for HC5, HC20, HC50 and HC80 are illustrative of some interesting properties of the SSD's derived through the MC sampling process. All plots show a very strong correlation between the HC values derived from the curve-fitting and the MC sampled SSDs. However, the SSD from the MC is systematically more conservative at the extremes

Active learning

Conclusion

Limitations

- no per-species uncertainty measurement
- little use of chemical and species information, making extending the model to untested chemicals impossible
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Acknowledgement

References

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